

SQL INTERVIEW PREP

TOP 20 CTE PATTERNS

Common Table Expressions • Modern SQL

1. Basic Readable CTE

2. Multiple CTEs

3. Recursion (1 to 10)

4. Manager Hierarchy

5. Nth Highest Salary

6. Delete Duplicates

7. Running Total

8. Moving Average

9. Gaps & Islands

10. Date Generation

11. YoY Growth (Lag)

12. Monthly Active Users

13. Find Missing IDs

14. String Splitting

15. Basket Analysis

16. Pivot Prep

17. Consecutive Wins

18. Update via CTE

19. Tree Path Generation

20. Insert from CTE

Q1

BASIC READABILITY

Replacing Nested Subqueries



Standard SQL

-- Don't nest. Define first.

```
WITH HighSales AS (  
    SELECT * FROM orders  
    WHERE amount > 1000  
)  
SELECT c.name, h.amount  
FROM customers c  
JOIN HighSales h ON c.id = h.cust_id;
```

-- Result: Much cleaner code!



Anandnarayanan S

Follow for more SQL Patterns

SQL CTE PREP

Q2

MULTIPLE CTEs

Chaining Logic Steps



Standard SQL

```
-- Separate by comma
WITH Sales2023 AS (
    SELECT * FROM sales WHERE year=2023
),
Sales2024 AS (
    SELECT * FROM sales WHERE year=2024
)
SELECT * FROM Sales2023 a
JOIN Sales2024 b ON a.prod_id = b.prod_id;
```



Anandnarayanan S

Follow for more SQL Patterns

SQL CTE PREP

Q3

RECURSION 101

Generating Numbers 1 to 10



SQL Server / PostgreSQL

```
-- The Base Case + Recursive Step
WITH RECURSIVE NumSeries AS (
  SELECT 1 AS n -- Anchor
  UNION ALL
  SELECT n + 1 -- Recursion
  FROM NumSeries
  WHERE n < 10 -- Terminator
)
SELECT * FROM NumSeries;
```



Anandnarayanan S

Follow for more SQL Patterns

SQL CTE PREP

Q4

MANAGER HIERARCHY

Who reports to whom?



Standard SQL

```
WITH RECURSIVE OrgChart AS (  
  SELECT id, name, manager_id, 1 as lvl  
  FROM employees WHERE manager_id IS NULL  
  UNION ALL  
  SELECT e.id, e.name, e.manager_id, o.lvl+1  
  FROM employees e  
  JOIN OrgChart o ON e.manager_id = o.id  
)  
SELECT * FROM OrgChart;
```



Anandnarayanan S

Follow for more SQL Patterns

SQL CTE PREP

Q5

2ND HIGHEST SALARY

Using Dense Rank + CTE



Standard SQL

```
-- Rank first, Filter second
WITH RankedSalaries AS (
  SELECT salary,
    DENSE_RANK() OVER(ORDER BY salary DESC) as rnk
  FROM employees
)
SELECT salary
FROM RankedSalaries
WHERE rnk = 2;
```



Anandnarayanan S

Follow for more SQL Patterns

SQL CTE PREP

Q6

DELETE DUPLICATES

Clean Data like a Pro



SQL Server / PostgreSQL

```
-- Identify dupes, then delete from CTE
WITH DupeRows AS (
  SELECT *,
    ROW_NUMBER() OVER(
      PARTITION BY email ORDER BY id
    ) as row_num
  FROM users
)
DELETE FROM DupeRows
WHERE row_num > 1;
```



Anandnarayanan S

Follow for more SQL Patterns

SQL CTE PREP

Q7

RUNNING TOTAL

Cumulative Sum



Standard SQL

```
-- Prepare data, then window
WITH DailySales AS (
  SELECT date, SUM(amount) as daily_amt
  FROM sales GROUP BY date
)
SELECT date, daily_amt,
  SUM(daily_amt) OVER(ORDER BY date) as run_tot
FROM DailySales;
```



Anandnarayanan S

Follow for more SQL Patterns

SQL CTE PREP

Q8

MOVING AVERAGE

3-Day Rolling Avg



Standard SQL

```
WITH DailyStats AS (  
  SELECT date, COUNT(*) as orders  
  FROM sales GROUP BY date  
)  
SELECT date, orders,  
  AVG(orders) OVER(  
    ORDER BY date  
    ROWS BETWEEN 2 PRECEDING AND CURRENT ROW  
  ) as 3_day_avg  
FROM DailyStats;
```



Anandnarayanan S

Follow for more SQL Patterns

SQL CTE PREP

Q9

GAPS + ISLANDS

Grouping Consecutive Dates



Standard SQL

```
-- Hardest Interview Question!  
WITH Grouped AS (  
  SELECT date,  
    -- Date minus Row Number creates constant  
    -- for consecutive sequences  
    DATEADD(day, -ROW_NUMBER() OVER(ORDER BY date), date) as grp  
  FROM active_days  
)  
SELECT MIN(date), MAX(date)  
FROM Grouped GROUP BY grp;
```



Anandnarayanan S

Follow for more SQL Patterns

SQL CTE PREP

Q10

DATE GENERATION

Fill Missing Dates



PostgreSQL

```
-- Generate Calendar Table on fly
WITH RECURSIVE Calendar AS (
  SELECT '2024-01-01'::date as dt
  UNION ALL
  SELECT dt + 1
  FROM Calendar
  WHERE dt < '2024-01-31'
)
SELECT c.dt, COALESCE(s.amount, 0)
FROM Calendar c
LEFT JOIN sales s ON c.dt = s.date;
```



Anandnarayanan S

Follow for more SQL Patterns

SQL CTE PREP

Q11

YOY GROWTH

Using LAG() in CTE



Standard SQL

```
WITH YearlySales AS (  
    SELECT YEAR(date) as yr, SUM(amt) as tot  
    FROM sales GROUP BY YEAR(date)  
)  
SELECT yr, tot,  
    LAG(tot) OVER(ORDER BY yr) as prev_yr,  
    (tot - LAG(tot) OVER(ORDER BY yr)) /  
    LAG(tot) OVER(ORDER BY yr) * 100 as growth  
FROM YearlySales;
```



Anandnarayanan S

Follow for more SQL Patterns

SQL CTE PREP

Q12

MONTHLY ACTIVE

Retention Analysis



Standard SQL

```
WITH MonthlyUsers AS (  
    SELECT DISTINCT user_id,  
        DATE_TRUNC('month', date) as mth  
    FROM logins  
)  
SELECT m1.mth,  
    COUNT(m1.user_id) as active_users,  
    COUNT(m2.user_id) as retained_next_mth  
FROM MonthlyUsers m1  
LEFT JOIN MonthlyUsers m2  
    ON m1.user_id = m2.user_id
```



Anandnarayanan S

Follow for more SQL Patterns

SQL CTE PREP

Q13

FIND MISSING IDS

Gaps in Sequence

Standard SQL

```
WITH RECURSIVE Expected AS (  
  SELECT 1 as id  
  UNION ALL SELECT id + 1 FROM Expected  
  WHERE id < (SELECT MAX(id) FROM users)  
)  
SELECT e.id  
FROM Expected e  
LEFT JOIN users u ON e.id = u.id  
WHERE u.id IS NULL;
```



Anandnarayanan S

Follow for more SQL Patterns

SQL CTE PREP

Q14

STRING SPLITTING

Parsing "Apple,Banana,Orange"



SQL Server

```
WITH RECURSIVE Split AS (  
    SELECT  
        LEFT(val, CHARINDEX(',', val+',')-1) as item,  
        STUFF(val, 1, CHARINDEX(',', val+','), '') as rest  
    FROM strings  
    UNION ALL  
    SELECT  
        LEFT(rest, CHARINDEX(',', rest+',')-1),  
        STUFF(rest, 1, CHARINDEX(',', rest+','), '')  
    FROM Split WHERE rest > ''  
)
```



Anandnarayanan S

Follow for more SQL Patterns

SQL CTE PREP

Q15

BASKET ANALYSIS

Products bought together



Standard SQL

```
WITH Pairs AS (  
  SELECT a.prod_id as p1, b.prod_id as p2  
  FROM order_items a  
  JOIN order_items b  
    ON a.order_id = b.order_id  
    AND a.prod_id < b.prod_id  
)  
SELECT p1, p2, COUNT(*) as frequency  
FROM Pairs  
GROUP BY p1, p2  
ORDER BY frequency DESC;
```



Anandnarayanan S

Follow for more SQL Patterns

SQL CTE PREP

Q16

PIVOT PREPARATION

Rows to Columns Logic



Standard SQL

```
WITH BaseData AS (  
    SELECT year, qtr, revenue FROM sales  
)  
SELECT year,  
    MAX(CASE WHEN qtr=1 THEN revenue END) as Q1,  
    MAX(CASE WHEN qtr=2 THEN revenue END) as Q2,  
    MAX(CASE WHEN qtr=3 THEN revenue END) as Q3  
FROM BaseData  
GROUP BY year;
```



Anandnarayanan S

Follow for more SQL Patterns

SQL CTE PREP

Q17

CONSECUTIVE WINS

Using Lead/Lag



Standard SQL

```
-- 3 wins in a row?  
WITH Streaks AS (  
  SELECT id, result,  
    LAG(result, 1) OVER(ORDER BY id) as prev1,  
    LAG(result, 2) OVER(ORDER BY id) as prev2  
  FROM matches  
)  
SELECT * FROM Streaks  
WHERE result='Win' AND prev1='Win' AND prev2='Win';
```



Anandnarayanan S

Follow for more SQL Patterns

SQL CTE PREP

Q18

UPDATE VIA CTE

Modifying Data Safely

SQL Server

```
-- Updates the underlying table
WITH ToUpdate AS (
    SELECT * FROM employees
    WHERE performance_score > 90
)
UPDATE ToUpdate
SET salary = salary * 1.10;
```



Anandnarayanan S

Follow for more SQL Patterns

SQL CTE PREP

Q19

TREE PATHS

Breadcrumbs: Home > Elec > TV

Standard SQL

```
WITH RECURSIVE Paths AS (  
  SELECT id, name, CAST(name AS VARCHAR) as path  
  FROM categories WHERE parent_id IS NULL  
  UNION ALL  
  SELECT c.id, c.name,  
         p.path || ' > ' || c.name  
  FROM categories c  
  JOIN Paths p ON c.parent_id = p.id  
)  
SELECT * FROM Paths;
```



Anandnarayanan S

Follow for more SQL Patterns

SQL CTE PREP

Q20

INSERT FROM CTE

ETL / Data Migration



Standard SQL

```
-- Calculate complex logic, then Insert
INSERT INTO summary_table
WITH CalculatedData AS (
    SELECT region, SUM(sales) as total
    FROM raw_sales
    GROUP BY region
)
SELECT * FROM CalculatedData;
```



Anandnarayanan S

Follow for more SQL Patterns

SQL CTE PREP

MISSION COMPLETE



Hi,
I'M ANAND

FOUND IT HELPFUL?

Help your network learn this too!



REPOST THIS CHEATSHEET



Anandnarayanan S

Follow for more SQL Patterns

SQL CTE PREP